

HydroRational User Manual

Version 0.9.0 beta, to the County of San Diego 2026 Hydrology Manual. Updated September 22, 2026.

1. About HydroRational

HydroRational computes rational method peak flows, Section 6 hydrographs and detention routing to the 2026 County of San Diego Hydrology Manual, then exports a plan check report. Open it at **hydrorational.web.app** in any current desktop browser. There is nothing to install.

Scope. San Diego County only. The C table, Ti table, intensity method and report format are the County's 2026 procedures, and no other jurisdiction is offered. Sites outside the county are flagged, and rainfall is not fetched for them.

Beta status. Version 0.9.0 is a beta. The hydrology is validated against the manual and against SDHydroTools (Tc 10 min, A 20 ac, C 0.60 gives Q 34.3 cfs, peak at 245 min, volume 49.30 ac-in), but the report has not yet been used on a live submittal. Results remain the responsibility of the engineer of record, as the report itself states.

What it covers

- Runoff coefficient from Table 3-1, initial time from Table 3-2, travel time by watercourse type (Section 3.1.4.2)
- Intensity by log-log interpolation of NOAA Atlas 14 depths (Section 3.1.3), fetched automatically for the project site
- Junction confluence by the Section 3.4 equation, with Notes 1, 2 and 3, accumulated node to node
- Section 6 six hour rational method hydrograph and Modified Puls detention routing
- Terrain from USGS 3DEP plus your own survey or grading surface, one click watershed delineation, and basin storage read from the ground
- Existing and proposed scenarios with a pre and post discharge comparison
- Report as PDF or text, drawing as DXF, project as JSON
- This manual and the 2026 County manual, both readable inside the app

2. The screen

The window has five areas: a toolbar across the top, the layer panel on the left, the map in the centre, the properties panel on the right, and a results drawer along the bottom.

Area	What it holds
Toolbar, file buttons	New project, Open project (JSON), Save project (JSON), Export text report, Export DXF, Export PDF report. Hover any icon for its name.
Toolbar, drawing tools	Select, Subcatchment, Junction / Node, Basin, Outfall, Link / Flow path
Toolbar, view buttons	User manual (?), Fit project to view, Fetch FEMA / NHD data for the current view, Toggle results drawer, dark mode
Toolbar, right side	Scenario selector with + Proposed, SD County 2026 (opens the County manual), Run Analysis , and the save status chip
Layer panel (left)	Project, Network layers, Terrain, Public datasets (3DEP, FEMA NFHL, NRCS soils, NHD, NWI wetlands, NLCD) and User layers. Click a group name to expand it.
Map (centre)	Address and coordinate search at the top. Zoom, My location and basemap buttons. The basemap bar at the bottom left switches between Esri Streets, Esri Topo, USGS Topo, Satellite and OpenTopo. The status bar shows the active tool, cursor position, zoom level and basemap.
Properties panel (right)	The workspace cards when nothing is selected, or the properties of whatever you click: subcatchment, node, flow path or basin
Results drawer (bottom)	Tabs for Hydrograph, Summary, Profile, Pre / Post and Report. Click the arrow to raise or lower it.

The save status chip reads **Local** when accounts are not active, **Saving...** while an upload runs, and **Saved to your account** once the cloud copy is current.

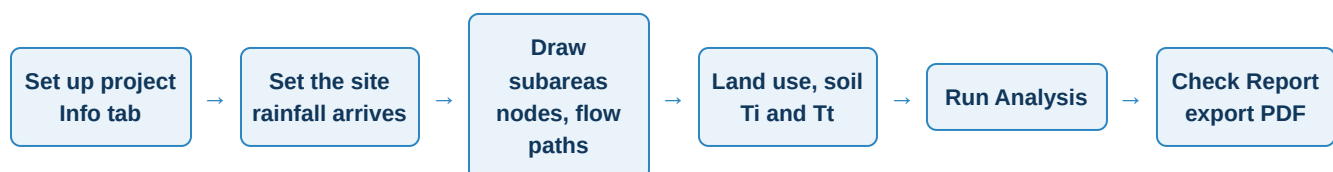
Help and manuals

Both manuals open in a side panel on the right, so you can read while you work. The map and panels stay usable, and you can drag the panel's left edge to resize it.

- **?** in the toolbar, or **F1**, opens this manual. **SD County 2026** opens the 2026 County Hydrology Manual. Each button closes its own manual, or switches to it if the other is showing.
- The two tabs at the top of the panel switch between manuals, and each remembers its page. The list beside them jumps to any section, table, figure or appendix.
- Citations in the app are links. Table 3-1, Section 3.4 and the other references in the properties panel open the County manual at that page. Help links such as **Read the quick start** open this manual.
- **New tab** opens the current page in its own browser tab, which is handy on a second screen. **Esc** or **Close** hides the panel without losing your place.

3. Quick start

A single subarea study takes about ten minutes: set up the project, get rainfall, draw, run, export.



1. With nothing selected, click **Set up project** in the properties panel. Enter the project name, site address or APN, and engineer of record, then **Save and continue**.
2. Set the project site. Search for the address in the box at the top of the map and click **Set as project site and fetch rainfall**, or type the latitude and longitude on the Info tab. On hydrorational.web.app the NOAA Atlas 14 depths fill in by themselves. Open the Rainfall tab to confirm them and pick the design frequency (100, 50, 25 or 10 year).
3. Press **P** (or click Subcatchment), click the polygon vertices on the map, and double-click to finish. The area computes automatically.
4. In the properties panel, choose the land use from Table 3-1 and the soil group, then enter the overland length and slope for Ti and the flow path length and slope for Tt. C, Ti, Tt, Tc, I and Qp update as you type.
5. Add an outfall with **O** and set the subarea's **Drain to node** to it. For larger systems add junctions with **J** and connect them with flow paths using **L**.
6. Click **Run Analysis** (or press **Enter**). Results fill the drawer at the bottom.
7. Check the Report tab, then export with the PDF or text icon in the toolbar.

The sample NOAA file PF_Depth_English_PDS.csv in the repository's samples folder is a quick way to try the workflow without a real site.

4. Project setup and rainfall

The project record and the NOAA Atlas 14 depths drive every intensity in the study, so set both before drawing. Open them from the workspace cards, or from **Project info** and **NOAA Atlas 14 rainfall** in the layer panel. The panel has three tabs: Info, Rainfall and Options.

Info tab

Field	Notes
Project name, Site address / APN, County / Agency, Date	Printed in the report header
Engineer of record: name and credentials, license no., firm	Printed in the report header and signature block
Site coordinates: latitude and longitude	The rainfall location. Typing both fetches rainfall for that point (see below). Center map on site moves the map there.

Rainfall tab

Study parameters. **Design frequency** is 100, 50, 25 or 10 year and selects which ARI column of the NOAA table is read. **Jurisdiction** is SD County 2026 and cannot be changed.

Depths. The app needs the 5, 10, 15, 30 and 60 minute and 2, 3 and 6 hour depths in inches. The 6 hour depth is P6 (P360) for the Section 6 hydrograph. There are five ways to fill the boxes.

1. **Automatically, when you set the site.** On hydrorational.web.app the server proxy is deployed. Setting the site from the map search, or typing both coordinates on the Info tab, fetches the NOAA Atlas 14 depths for that point.

2. **Auto-fetch NOAA Atlas 14** on the Options tab. Fetches for the site coordinates, or for the centre of the map if none are set. Use it to replace rainfall you entered yourself.
3. **Load NOAA CSV**. Click **Get CSV from NOAA** to open the PFDS page for the site, download the precipitation frequency estimates CSV (depth, English units, partial duration series), and load it. Latitude, longitude and location name are read from the file.
4. Drag the CSV onto the depth boxes.
5. Paste the NOAA table into any depth box.

One CSV covers every return period. Change the design frequency and the matching column is used. Click **Save rainfall data** when the depths are in.

Rainfall you entered is never replaced

The app records where the depths came from, fetched or entered by you, and saves that with the project.

When you	What happens to the depths
Set or move the site, depths were fetched	Fetched again for the new site
Set or move the site, depths were typed, pasted or loaded from a CSV	Kept. A message says so. Use Auto-fetch NOAA Atlas 14 if you do want to replace them.
Change the design frequency, depths were fetched	Fetched again for the new frequency
Change the design frequency, depths came from a CSV	The matching column of the loaded file is used
Set a site outside San Diego County	Nothing is fetched, and a message says the site is outside the county
Open a project saved before this feature	Its depths are treated as entered by you, so they are never replaced

Without the proxy (for example when the app is opened as a local file), nothing is fetched automatically. The map search then offers **Open NOAA Atlas 14 for here**, and the CSV is the way in.

Options tab

Option	Default	Effect in 0.9.0
Server proxy URL	This site	Enables fetching of NOAA and SSURGO data. Leave as is on the hosted app. Use this site restores it.
Auto-fetch NOAA Atlas 14	Button	Fetches depths for the site coordinates, replacing whatever is in the boxes
Use Note 3 effective Tc at junctions	On	Changes the junction result, see section 7
Min pipe size (in), Friction slope factor, Output format	18, 0.90, Node and station listing	Placeholders. They are not yet read by the calculations or the report.

5. Building the drainage network

A study is subareas that drain to nodes, and nodes joined by flow paths down to an outfall. Draw them in that order: subareas, then nodes, then the links between nodes.

Element	Tool	Key	How to place it
Subcatchment	Subcatchment	P	Click vertices, double-click to finish. Area is computed from the polygon.
Junction / node	Junction / Node	J	Click once on the map
Detention basin	Basin	B	Click once. A basin is a node with outlet and storage data.
Outfall	Outfall	O	Click once at the point of discharge
Flow path	Link / Flow path	L	Draw from the upstream node to the downstream node, clicking bends along the way, double-click to finish

Press **Esc** or **S** to return to Select, then click any element to open its properties. **Delete** (or **Backspace**) removes the selected element.

Connecting subareas. In each subarea's Identity section set **Drain to node**. If you also set **Upstream node**, travel time is summed along the flow paths between the two. A subarea with no drain-to node is computed on its own but does not reach any junction.

Station numbers. Enter **Station from** and **Station to** for each subarea and a node ID for each node. These label the blocks in the node and station listing.

Importing GIS data instead of drawing

Under User layers, click **Import shapefile, GeoJSON, KML or KMZ**, or drop the file on the map. Shapefiles go in as a .zip holding the .shp, .dbf and .prj. California State Plane zones 5 and 6 are reprojected automatically. Once a polygon layer is loaded, click **Make subcatchments** on its row to turn every polygon into a subarea.

Reference layers

The Public datasets group shows FEMA NFHL flood zones, NRCS soils by hydrologic soil group, NHD streams and water bodies, NWI wetlands and NLCD 2021 land cover. Toggle a layer on, then fetch it for the current view. Basemaps are switched from the bar at the bottom left of the map. These layers are for reference and do not change any computed value unless you use one of the sampling buttons in the subarea panel.

6. Subcatchment properties

Select a subarea and the properties panel computes C, Ti, Tt, Tc, I and Qp live as you type. The panel has three tabs: Properties, Tc / Flow and Hydrograph. The section and table numbers printed in the panel are links into the County manual.

Runoff coefficient (Table 3-1, Section 3.1.2)

Choose the land use and the NRCS hydrologic soil group. Land use sets the default percent impervious, which you can overwrite.

$$C = 0.90 \times I_{\text{imp}} + C_p \times (1 - I_{\text{imp}})$$

Here I_{imp} is the impervious fraction and C_p is the pervious coefficient: A 0.20, B 0.25, C 0.30, D 0.35. C is capped at 0.95. The panel also reports the weighted CA.

Land use (Table 3-1)	Default % impervious
Natural / Permanent Open Space	0
LDR Residential 1.0 DU/A	10
LDR Residential 2.0 DU/A	20
MDR Residential 4.3 DU/A	35
MDR Residential 7.3 DU/A	40
MDR Residential 10.9 DU/A	50
HDR Residential 24 DU/A	65
Neighborhood Commercial	80
General Commercial	85
General Industrial	95

Two helpers fill these inputs from public data:

- **Get soil group from NRCS SSURGO** looks up the HSG under the subarea through the server proxy.
- **Sample percent impervious from NLCD** averages about thirty points across the subarea and reports the point count and spread with the mean. It is a sample, not an exact area weighting, so check it against the site plan.

Initial time T_i (Table 3-2)

Enter the overland length and slope. T_i is interpolated from Table 3-2 by land use and slope (0.5 to 10 percent), with a minimum of 5 minutes. The panel shows the maximum overland length L_m for that row. Keep the overland length at or below L_m and carry the remainder as travel time.

Travel time T_t (Section 3.1.4.2)

Pick the flow path method, Urban gutter / street (Manning, $n = 0.013$) or Natural channel (Kirpich), then the length and slope. The panel shows velocity and T_t . If the subarea has both Upstream node and Drain to node set, T_t is instead summed along the drawn flow paths between them.

Peak flow (Section 3.1.1)

$$Q_p = C \cdot I \cdot A \quad T_c = T_i + T_t$$

I is read at T_c from the NOAA Atlas 14 depths by log-log interpolation (Section 3.1.3). If I shows a dash, the rainfall depths have not been entered. **Recalculate** forces a refresh.

Generate hydrograph on the Hydrograph tab builds the Section 6 hydrograph for this subarea alone.

7. Nodes, junctions and flow paths

Nodes collect flow, and flow paths carry it downstream with a travel time. Peak flows accumulate node to node by the Section 3.4 junction equation.

Node properties

- **Node ID** and **Type**: Junction, Detention basin, or Outfall / discharge point. Changing the type to Detention basin enables the basin panel (section 10).
- **Elevation (ft)**: type it, or click **Get from DEM / USGS**. That uses a loaded terrain surface if there is one, otherwise USGS 3DEP. **All nodes** fills every node at once.
- **Flow paths at this node** and **Subareas draining here** list what is connected, which is the quickest way to find a missing link.
- **MRM junction result** shows the controlling Tc and the Qp leaving the node after a run.

Flow path properties (Section 3.1.4.2)

Set the From and To nodes, the watercourse type, the length and the slope. **Slope from node elevations** computes the slope from the two node elevations. Tt is length divided by velocity, and segment times add along the route. The Manning n can be overwritten on each flow path.

Watercourse type	Method	Default n
Street gutter	Manning	0.013
Pipe / conduit	Manning, pipe diameter entered	0.013
Swale / ditch	Manning	0.030
Improved channel	Manning	0.020
Natural channel	Kirpich	none

Junction analysis (Section 3.4)

When two or more streams meet, the app sorts them by Tc and evaluates the junction equation at each stream's Tc. A stream with a shorter Tc is scaled by the intensity ratio, a stream with a longer Tc by the Tc ratio, and the largest total governs. The three notes of Section 3.4 apply:

1. **Note 1**, equal Tc: flows add directly.
2. **Note 2**, Tc values within 10 percent: $Q = \Sigma(CA) \times I$ at the shorter Tc.
3. **Note 3**, effective Tc: on by default, and switched on the Options tab. The app backs the intensity out of the combined Q and reads the matching Tc, so downstream flows do not fluctuate.

The manual prints the Note 3 relation as $I = \Sigma(CA)/Q$. Because $Q = \Sigma(CA) \times I$, the app uses $I = Q/\Sigma(CA)$. Confirm this reading with the reviewing agency before a submittal relies on Note 3, or turn Note 3 off.

8. Terrain and watershed delineation

With terrain loaded, one click at an outlet produces a subarea with its boundary, area, overland reach and watercourse measured from the ground. Everything is in the Terrain group of the layer panel.

Load the terrain

1. Zoom to the site and click **Fetch 3DEP for this view**. This is the base surface for the whole tributary area.
2. To add your own surface, set the four boxes first, then click **Load survey or proposed surface** and choose a GeoTIFF, ESRI ASCII grid or x,y,z point file.
 - Horizontal system: CA State Plane 6 (ft), CA State Plane 5 (ft), or lat / long WGS84. San Diego County is zone 6.
 - Vertical unit: feet or metres.
 - Datum shift (ft): added to every elevation, for example to move a survey onto NAVD88.
 - Layer: Existing survey or Proposed grading. Proposed sits above existing, which sits above 3DEP.
3. Each surface appears as a row under the buttons. Higher priority surfaces win inside their footprint and 3DEP fills the rest.

The file cannot tell the app its coordinate system or vertical unit, so those boxes matter. A wrong vertical unit makes the ground read 3.28 times too high or too low.

Check the seam before delineating

Where a survey meets 3DEP there is a vertical seam. A mismatch there creates a false ridge or sink that pushes flow the wrong way. On loading, the app reports the mean and worst difference and warns when it exceeds 0.5 ft. Turn on **Seam check** to draw the boundary cells, and correct the datum shift if the seam is large.

Terrain tools

Tool	Input	Use
Contours	Interval in ft (default 5)	Toggle on to draw contours from the composited surface
Draw a terrain profile	Draw a line, double-click to finish	Length, fall and grade, plotted in the Profile tab
Channels	Contributing area in acres (default 1)	Draws the flow network above that threshold. Use it to confirm the surface routes sensibly before trusting a delineation.

Delineate a watershed

1. Click **Delineate a watershed**, then click the outlet. The point snaps to the nearest channel.
2. The app fills depressions, routes flow, and draws the contributing boundary as a new subarea with its area.
3. It traces the hydraulically longest flow path and splits it into an initial overland reach and the watercourse below it, each with its own length and slope. These land in the Ti and Tt inputs. The two reaches are drawn in different colours so the numbers can be checked on the map.
4. Set land use and soil as usual.

If the message says the point drains almost nothing, click further down the slope. **Edit boundaries** lets you drag the vertices of any subarea.

Provenance: your entries are never overwritten

Every terrain derived value records where it came from. The subarea panel states it, for example *boundary from terrain, 4 values from terrain, 1 overridden*. Once you type over a value or edit a boundary, that field is yours.

Re-run delineation, keeping my edits appears on delineated subareas. It applies fresh terrain values only to fields you have not touched, which is the step to use after loading a revised surface. Provenance is saved with the project.

9. Existing and proposed scenarios

A drainage study compares two conditions, and the Pre / Post tab gives the peak discharge change at every point of discharge.

1. Build and run existing conditions first.
2. Click **+ Proposed** in the toolbar. This copies the existing scenario, so you only draw what changes.
3. With Proposed selected in the scenario dropdown, load the graded surface as Proposed grading, then use **Re-run delineation, keeping my edits** on affected subareas, or redraw them.
4. Update land use, percent impervious, flow paths and any basins, then Run Analysis.
5. Open the Pre / Post tab.

Each scenario keeps its own subareas, nodes, flow paths and terrain stack. Terrain rasters are stored once and shared, so the file does not double in size.

The comparison matches outfalls by node, and reports existing Q, proposed Q, the change and the percent change. A point that exists in only one condition is flagged rather than dropped, so give matching outfalls the same node ID in both scenarios. The same table goes into the report, which also states which condition the node and station listing describes.

10. Detention basins

A basin routes the Section 6 inflow hydrograph through its outlet by Modified Puls storage indication, with the routing step equal to T_c . Place it with **B**, or set an existing node's type to Detention basin, then click **Open basin outlet and stage-storage**. At least one subarea must drain to the basin.

Outlet tab

Device	Inputs	Equation
Low flow orifice	Diameter (in), invert elevation (ft), discharge coefficient Cd	$Q = C_d A \sqrt{2gh}$, continuous from part full through submerged
Emergency overflow weir	Weir length (ft), crest elevation (ft), weir coefficient Cw	$Q = C_w L H^{1.5}$

Stage-Storage tab

Enter cumulative storage in acre-feet against elevation with **Add row**. Storage is interpolated linearly between rows. For a graded basin, let the terrain do it:

1. Load the proposed grading surface (section 8).
2. Click **Read storage from the terrain**. Set the level step (ft) and the maximum depth to fill (ft) in the two boxes beside it.

3. The app floods the depression at the basin node from its low point and returns storage against elevation, the floor, and the spill elevation.
4. Click **Set the weir crest to the spill elevation** to put the emergency weir where the basin actually overtops.

The fill stops at the real rim. If water would cross a saddle into lower ground, that saddle is the spill. If the depression runs off the loaded terrain, or the depth limit is reached first, the app says so instead of reporting a rim that is not there. Any row you edit afterwards is kept.

Routing tab

Click **Route the inflow hydrograph**. Results: peak inflow and outflow (cfs), peak reduction (%), maximum water surface (ft), storage used (ac-ft), freeboard (ft), time to peak outflow and the routing step.

Two warnings to act on: *Water surface rises above the top of the stage-storage table* means the table must be extended, and *Emergency weir is active at the peak* means the design storm overtops to the spillway.

11. Running the analysis and reading results

Network flows recompute every time you edit a subarea, node or flow path, so node results are always current.

Run Analysis (or **Enter**) builds the Section 6 hydrograph and summary for the selected subarea, or the first one if none is selected, and opens the results drawer.

Before running, the app checks three things and stops with a message if one is missing: NOAA depths entered, at least one subarea drawn, and a computable Qp. It also warns when subareas do not drain to any node, because junction confluence is not computed for them until they do.

Tab	Shows
Hydrograph	The Section 6 six hour nested storm hydrograph for the subarea, in blocks of length Tc
Summary	Project, subarea, area, land use, soil, C, CA, Ti, Tt, Tc, I, Qp, number of hydrograph blocks, peak time and total volume in ac-in
Profile	The last terrain profile drawn (section 8)
Pre / Post	Discharge comparison, visible once a Proposed scenario exists (section 9)
Report	Preview of the node and station listing and summary tables that the PDF and text exports contain

To check a single subarea without running, open its Hydrograph tab in the properties panel and click **Generate hydrograph**.

Self-check against the SDHydroTools benchmark. Enter depths of 0.332, 0.476, 0.576, 0.790, 1.119, 1.518, 1.815 and 4.108 in (5 min through 6 hr), and a subarea with Tc 10 min, A 20 ac and C 0.60 (Qp 34.3 cfs). The hydrograph has 36 blocks, a peak at 245 min and a volume of 49.30 ac-in.

12. Reports and exports

All exports are the icon buttons at the left of the toolbar. Files are named from the project name, or the site address if there is no name. Preview the report in the Report tab before exporting.

Export	File	Contents
PDF report	<project>_Hydrology_Report.pdf	Project block with engineer and site coordinates, NOAA Atlas 14 depths, subarea summary, flow path summary, node and station listing, Section 6 hydrograph, pre and post comparison when two scenarios exist
Text report	<project>_Node_Station_Report.txt	The node and station listing in plain text, for pasting into a study or comparing against another program
DXF drawing	<project>_drainage_map.dxf	AutoCAD R12 in the working coordinate system, on layers HR-SUBAREA, HR-FLOWPATH, HR-NODE, HR-POND, HR-CONTOUR (if contours are on) and HR-TEXT
Project	.json	The whole study, see section 13

The DXF is written in the project's working coordinate system (California State Plane when the terrain is in State Plane), so it drops onto the survey in CAD without moving or scaling. The status bar names the system after each export.

The node and station listing follows the block layout San Diego plan checkers are used to, not a published specification. Until it has been compared against a run you trust, read the first few reports side by side with your usual program. The sample export docs/sample_report.pdf in the repository shows the format.

13. Saving, accounts and cloud sync

On hydrorational.web.app you are signed in as a guest the moment the page opens, and your work saves to the cloud about fifteen seconds after each change. No sign-up is needed to start.

Keep your projects across browsers

1. Open the Account panel, enter your email, and click **Email me a sign-in link**.
2. Open the link in the same browser. The address is linked to the same account, so your guest projects are still there.
3. On another computer, request a link to the same address. Projects from that browser are copied into the account.

There is no password, and the email is used only for the sign-in link. **Your projects** lists saved studies. **Save now** forces an upload, **New project** starts a blank study, and **Sign out** ends the session. Each user can read and write only their own projects.

Project files

Save project (JSON) downloads the whole study, and **Open project (JSON)** loads one back. You can also drop a .json on the map. Keep a JSON copy with each submittal, since it is the reproducible record of the study.

The file holds both scenarios, all inputs and their provenance, including where the rainfall came from. Survey and proposed surfaces are stored inside it, rounded to 0.01 ft, because they cannot be re-fetched. 3DEP is stored as a footprint only and re-fetched on demand, which keeps the file small.

Opened as a local file rather than from the website, the app runs without accounts or automatic rainfall, and saving is by JSON download only.

14. Reference

Keyboard shortcuts

Key	Action
S	Select
P	Draw subcatchment
J	Place junction or node
B	Place detention basin
O	Place outfall
L	Draw link or flow path
Delete or Backspace	Remove the selected element
Enter	Run Analysis
Esc	Back to Select, and close the manual panel if it is open
F1	Open this manual

Letter keys and Enter are ignored while you are typing in a box.

Files the app accepts

Data	Formats	Where
Rainfall	NOAA PFDS CSV, or the table pasted	Rainfall tab, or drop on the depth boxes
Subarea and site polygons	Shapefile (.zip with shp, dbf, prj), GeoJSON, KML, KMZ	User layers, or drop on the map
Terrain	GeoTIFF, ESRI ASCII grid (.asc), x,y,z points	Terrain group
Project	HydroRational .json	Open project, or drop on the map

Manual references used

Each of these opens in the County manual from the jump list in the manual panel.

Computation	2026 Manual reference
Q = CIA	Section 3.1.1
Runoff coefficient	Section 3.1.2, Table 3-1
Rainfall intensity	Section 3.1.3, NOAA Atlas 14, log-log interpolation
Rainfall data	Appendix B
Initial time T_i	Table 3-2
Travel time T_t	Section 3.1.4.2
Junction confluence	Section 3.4, Notes 1 to 3
Rational method hydrograph	Section 6
Detention routing	Modified Puls storage indication

15. Troubleshooting and known limits

Symptom	Cause and fix
I and Qp show a dash	No rainfall depths. Set the site, or load the NOAA CSV on the Rainfall tab.
Run Analysis says enter NOAA depths first	Same as above. The app opens the Rainfall tab for you.
Rainfall did not change when I moved the site	The depths were typed, pasted or loaded from a CSV, so they were kept. Click Auto-fetch NOAA Atlas 14 on the Options tab to replace them.
Message that the site is outside San Diego County	Rainfall is not fetched outside the county. Check the coordinates, since a swapped sign or a mistyped search is the usual cause.
Auto-fetch NOAA or SSURGO does nothing	The server proxy is unreachable. Click Use this site on the Options tab, or fall back to the CSV and manual soil entry.
Message that subareas do not drain to a node	Set Drain to node on each subarea. Junction flows are not computed for loose subareas.
Ground elevations about 3.3 times too low or too high	Wrong vertical unit on a loaded surface. 3DEP is always metres and the app handles that. Reload your file with the correct unit.
Delineated boundary cuts across an obvious ridge	Check the seam report and Seam check. Correct the datum shift, then re-run delineation.
Delineation says the point drains almost nothing	Click further downstream, on a channel shown by Channels.
Basin warning: water rises above the stage-storage table	Extend the table, or read storage from the terrain with a larger maximum depth.
Pre / Post tab is missing	It appears only once a Proposed scenario exists.
Outfall flagged as present in one condition only	The node IDs differ between scenarios. Give matching outfalls the same ID.
Manual panel is blank	The browser's built-in PDF viewer is turned off, or you are on a phone. Click New tab to open the page directly.

Known limits in 0.9.0 beta

- The node and station listing has not yet been verified against another program or used on a live submittal.
- Note 3 uses $I = Q/\Sigma(CA)$, the algebraic inverse of $Q = \Sigma(CA) \times I$, where the manual prints $I = \Sigma(CA)/Q$. Note 3 is on by default.
- The San Diego County check on the project site uses the county's bounding box, not its boundary. It catches a site in the wrong state or county, but lets through slivers of Orange, Riverside and Imperial counties and Baja California.
- NLCD percent impervious is a point sample of about thirty points, not an area weighting.
- Min pipe size, friction slope factor and output format on the Options tab are not yet connected to the calculations.
- Not built: FEMA base flood elevation lines, LAS and DXF terrain import, and terrain editing inside the app.

San Diego County only. Results remain the responsibility of the engineer of record.